THE HALTING PROBLEM OF THE MACHINE $1RB1LA_0RC0RD_1LC1LA_1RE1RD_1LF0LA_{-}{-}{-}0LE$: A CONDITIONAL NON-HALTING THEOREM AND THE RECORD-TIMETABLE OBSTRUCTION

ABSTRACT. We give a complete structural analysis of the 6-state, 2-symbol Turing machine $M =$ `1RB1LA_0RC0RD_1LC1LA_1RE1RD_1LF0LA_---_0LE` and prove, conditional on a single precisely-stated computational assumption (master closure, Assumption M), that it never halts from the blank tape; unconditionally, non-halting holds over all runs whose round classes stay in the exhaustively verified range ($h \le 2000$, sampled to $1.2 \cdot 10^7$) — a doubly-exponential horizon whose step-count scaling is labeled as empirical (Proposition on the envelope). The analysis proceeds by a macroscopic phase calculus between visits of state D to blank cells, a closed family of recurrent anchors, a halving-chain prologue, and a *round law* proved by strong induction on the 2-adic class $h = \nu_2(\alpha + 1)$ of the leading counter, via a self-similar round decomposition (halving cascade, universal ladder, master segment). Magnitude decisions are handled unconditionally by a dominance inequality (stated as Lemma D of the companion derivation paper); the halting branch is structurally unreachable for $h \ge 82$ and perturbation-robust; $\Delta t(h) \in [2, 9]$ on all verified classes. The single remaining assumption is shown to be equivalent, by a finite-state reduction, to the continuation of a record timetable of the autonomous 2-adic walk, whose 32-bit slice is certified acyclic by a finite experiment (with non-eventual-periodicity beyond the slice stated as a conjecture) — the precise, irreducible core of the problem. All other components are proved algebraically or verified exhaustively over declared finite domains; every claim is classified in a ledger (algebraic / exhaustive / assumption-dependent).

1. INTRODUCTION

Let M denote the Turing machine with states $A, \ldots, F$ and transition table of Table 1, in the standard notation where, e.g., `1RB` means: write 1, move right, go to state B, and where the single undefined transition (state F reading 0) is the halting transition. The machine is one of the remaining undecided instances of the 6-state Busy Beaver problem; to our knowledge no analysis of it has been published.

	read 0	read 1
A	`1RB`	`1LA`
B	`0RC`	`0RD`
C	`1LC`	`1LA`
D	`1RE`	`1RD`
E	`1LF`	`0LA`
F	HALT	`0LE`

TABLE 1. The transition table of M.

Theorem 1.1 (Main, conditional form)**.** *M, started on the blank tape, never halts, conditional on Assumption 4.10 (master closure for all classes).*

Date: October 2026.

Proposition 1.2 (unconditional envelope, verified range; empirical status made explicit)**.** *Unconditionally, M does not halt within any run whose round classes stay in the exhaustively verified range (all classes $h \le 2000$; sampled agreement through $1.2 \cdot 10^7$) — in step-count terms, non-halting is proven over a horizon that is doubly exponential in the class index (below $2^{2^{2000}}$ in the record-climb heuristic).*

Status of the envelope. The finite part is unconditional: over the verified range the round table is single-path and carries no halting block (Theorem 4.3, computational components of §8); the anchors climb by $n' \ge n + 6$ per exit, and the record timetable of the residual climbs by a measured factor ≈ 4 per class level, so the first arrival of class h requires a number of rounds growing like 4^h, each of supercubic cost in the anchor index — hence the doubly-exponential horizon. The record climb, however, is a *measured* law of the autonomous walk (Theorem 8.1), not an unconditional theorem; we therefore state the envelope as a proposition whose step-count translation carries this explicitly labeled empirical component. A paper proof of an explicit lower bound $T(h) \ge 2^{c \cdot 2^h}$ (assembling $A_j \le \tau_j^2$ with the band $\Delta\alpha \le 2h + 6$) is left open. □

The proof occupies Sections 2–7. Its skeleton:

(1) (§2) A phase calculus for the instants when the head, in state D, reads a 0. Between two such instants the machine executes a fully determined finite block of elementary steps, and the possible tape shapes at these instants form a finite grammar with nine transition rules (Lemma 2.3). The unique halting path passes through a single grammar configuration $G(3, \cdot)$ (Lemma 2.4).

(2) (§3) The blank-tape run reaches at step 54 an *anchor* $P(3)$, and every anchor $P(n)$ is carried, by an explicit halving chain of length $m = \nu_2(n + 3)$, into a standard family $\Phi(w, m)$, $n + 3 = 2^m w$, w odd (Theorem 3.4).

(3) (§4–4.2) The Φ-trajectory visits *landmarks* $L(\alpha, t; s)$ indexed by a leading counter $\alpha \equiv 1 \pmod 6$ and a time parameter t; the evolution of (α, t) depends only on the class $h = \nu_2(\alpha + 1)$ (Theorem 4.3, the *round law*), proved by strong induction on h via a halving cascade (Lemma 4.4), a universal ladder (Lemma 4.8), and a uniform master segment (Assumption 4.10, uniformity clause — Remark 4.11).

(4) (§5–7) Exhausted-tail configurations exit to the next anchor with $n' \ge n + 6$ and $n' \equiv 3 \pmod 6$ (Theorem 5.1), and no reachable configuration is the halting one (Theorem 6.1). The anchors therefore climb forever.

Throughout, tape contents are described by *runs* (maximal blocks of 1s) separated by *gaps* (blocks of 0s); all configurations live on the right-infinite half of the tape beyond a fixed left sentinel, which the machine never crosses (this follows from the left-moving transitions returning before the sentinel; it is confirmed by every invariant below).

2. The phase calculus

2.1. Events and configurations.

Definition 2.1 (event)**.** An *event* is an instant at which the head, in state D, reads a 0. Let $\mathcal{C}$ denote the set of tape shapes (with head position) that can hold at an event.

Definition 2.2 (grammar)**.** Write a configuration as a list of runs $(r_1, r_2, \dots, r_k)$, $k \ge 2$, with specified gaps between them and a marked head cell (the 0 being read). The following four types exhaust the configurations that arise on the orbits considered:

- $R(a, b)$: runs (a, b), gaps $[1]$, head on the fresh 0 *immediately right* of the right run;
- $S_k(x, y, z, \dots)$: $k \ge 3$ runs, gaps $[1, 2, 1, 2, \dots]$, head on the *left* 0 of the 2-gap between runs 2 and 3;

- $G_k(a, b, \dots)$: $k \geq 2$ runs, gaps $[3, 1, 1, \dots]$, head on the *middle* 0 of the 3-gap between runs 1 and 2;
- $L_k(r_1, r_2, r_3, \dots)$: $k \geq 3$ runs, gaps $[1, 1, 1, \dots]$, head on the single-0 gap between runs 2 and 3.

(Thus L_2 is identified with R: both have the head just right of the block.)

Lemma 2.3 (phase rules)**.** *Between consecutive events, M executes exactly the following transitions, with the stated exact step counts:*

(R)	$R(a, b) \to S_3(a{+}1,\ b{-}2,\ 1)$	$2b + 3, \quad b \geq 3;$
(S≥)	$S_k(\dots, x, y, z, \dots) \to S_k(\dots, x{+}1,\ y{-}2,\ z{+}1, \dots)$	$2y + 3, \quad y \geq 3;$
(S2)	$S_k(\dots, x, 2, z, \dots) \to G_{k-1}(\dots, x{+}1,\ z{+}1, \dots)$	$7;$
(S1)	$S_k(\dots, x, 1, z, \dots) \to L_{k-1}(\dots, 1,\ x{+}z{+}3, \dots)$	$2x + z + 14;$
(G≥)	$G_k(a, b, \dots) \to S_{k+2}(1,\ a{-}3,\ 1,\ b{+}1, \dots)$	$2a + 3, \quad a \geq 4;$
(G3)	$G_k(3, b, \dots) \to G_{k+1}(1,\ 1,\ b{+}1, \dots)$	$9;$
(G2)	$G_k(2, b, \dots) \to L_{k+1}(1,\ 3,\ b{+}1, \dots)$	$16;$
(G1)	$G_k(1, b, \dots) \to \mathit{HALT}$	$6;$
(L≥)	$L_k(r_1, r_2, r_3, \dots) \to L_k(r_1{+}1,\ r_2,\ r_3{-}1, \dots)$	$2r_2 + 5, \quad r_3 \geq 2;$
(L1)	$L_k(r_1, r_2, 1, \dots) \to R/S_{k-1}(r_1{+}1,\ r_2, \dots)$	$2r_2 + 5.$

In (S1) the result is R when $k = 3$ and L_{k-1} when $k \geq 4$; in (L1) it is R when $k = 3$ and S_{k-1} when $k \geq 4$. The halting block (G1) ends with F reading 0; the block (G3) is an ordinary event block whose target $G(1, \cdot)$ halts in the next block.

Proof sketch. Each rule is obtained by executing the transition table from the stated configuration until the next event. We illustrate rule (R) for $R(3, 3)$ (tape $\dots 1^3\,0\,1^3\,0^\infty$, head on the fresh 0, state D; the trace below is a step-level recording of the machine):

t	state/head	effect
0	D@7	tentative write 1, right, E
1	E@8	write 1, left, F
2	F@7	*erase* (`0LE`), `left`, E
3	E@6	*erase* (`0LA`), `left`, A
4, 5	A@5, A@4	march left through the right run
6	A@3	read the gap 0: *fill* it (`1RB`), `right`, B
7	B@4	write 0 (`0RD`), `right`, D
8	D@5	march right writing 1s
9	D@6	next event: $S_3(4, 1, 1)$

The net effect is a five-phase mechanism: (i) D's tentative extension of the right run is *undone* by the E–F–E–A recoil; (ii) A marches left across the right run; (iii) A fills the inter-run gap with 1, growing the first run to length $a{+}1$; (iv) B zeroes one cell of the old right run, creating the new 2-gap; (v) D marches right, rebuilding, so that the old right run of length b leaves a residue $b-2$ to the left of the new gap and a fresh run of length 1 to its right: $S_3(a+1, b-2, 1)$, at total cost $2b+3$. The remaining rules follow by the same kind of shuttling; each was derived stepwise from Table 1. Rule-level faithfulness is machine-checked in two complementary ways: (i) the rule checker `rulecheck.mjs` compares every rule against the raw machine on parameter grids ($a \in \{1, \dots, 20\}$, $b \in \{2, 3, 5, 10\}$, tails of length 0–2): 564/564 rows agree exactly (this includes the boundary cases $y \in \{1, 2\}$, $a \in \{1, 2, 3\}$, and the halting case (G1)); (ii) trajectory-level comparison of the phase engine against the raw oracle over 956 anchor trajectories ($n = 2..957$): 955 agree field-by-field, including 16 halting anchors with exact halt times; the single exception

is $R(2,2)$, whose successor ($b = 2 < 3$) lies outside the declared grammar and is flagged, not mis-computed (direct simulation halts at +22). An earlier version of the table carried a folded (G3)→ HALT row and an incorrect (G2) image $L(1, b{+}1, b{+}1)$; against that version the checker reported 531 ok / 21 bad, the bad rows being precisely (G2)/(G3) — the correction here follows the raw-machine side of that checker. □

Lemma 2.4 (halting criterion)**.** *M halts from an event configuration iff the phase trajectory reaches $G_k(1, b, \dots)$ for some b and tail; equivalently, iff it reaches a G-state with leading head $a \in \{1, 3\}$ (rule (G3) carries $G(3, \cdot)$ to $G(1, \cdot)$ in one block). On every orbit considered below, the leading heads of all G-states are ≥ 2 and $\neq 3$, so no other rule can produce a halt.*

Proof. The only undefined transition of Table 1 is F reading 0. An exhaustive check of the ten rules shows the state F reads a 0 in the interior of an event block in exactly one grammar situation, namely the configuration $G(1, b, \dots)$ (rule (G1)); in all other blocks F reads 1s or the block ends at the next event first. Rule (G3) is the sole in-rule source of $G(1, \cdot)$. □

2.2. **Sweeps.** For efficiency we also use *swept* versions of the rules: consecutive applications of (S≥) with the same x-growth collapse into one step with closed-form cost. If y is odd, $j = (y-1)/2$ applications carry $(x, y, z) \mapsto (x + j,\ 1,\ z + j)$ at total cost $\sum_{i=0}^{j-1}(2(y - 2i) + 3)$; if y is even, $j = (y-2)/2$ applications carry $(x, y, z) \mapsto (x + j,\ 2,\ z + j)$. These identities are elementary and are used silently below.

3. Anchors and the prologue

Definition 3.1 (anchor)**.** For $n \geq 2$ let $P(n)$ denote the configuration $0^\infty 1^n 0 1^n 0^\infty$ with the head in state D on the fresh 0 immediately right of the right run; in grammar terms $P(n) = R(n, n)$.

Lemma 3.2 (entry)**.** *Started on the blank tape, M reaches $P(3)$ at step 54.*

Proof. Direct simulation of the 54 steps: the machine builds the pattern $1^3 0 1^3$ by the *A-B-C-D-E-F* loop and lands in state D on the fresh 0. (Step-level trace in the repository; verified by two independent simulators.) □

Definition 3.3 (the family Φ)**.** For odd $w \geq 3$ and $m \geq 1$:

$$\Phi(w, m) \;=\; G\big(w - 1,\ w - 1,\ 2w,\ 4w,\ \dots,\ 2^{m-1}w,\ 2^m w - 1\big).$$

Theorem 3.4 (prologue)**.** *Let $n \equiv 3 \pmod 6$, $n \geq 3$ (this holds for every anchor on the orbit of Theorem 1.1; in particular n is odd), and write $n + 3 = 2^m w$ with w odd, $m \geq 1$. Then the phase trajectory from $P(n)$ reaches $\Phi(w, m)$; explicitly,*

$$R(n, n) \;\to\; R(1,\ 2n + 2) \;\to\; G(n + 2,\ n + 1) \;\to^{m \text{ halvings}}\; \Phi(w, m),$$

where the halving steps act on the two leading runs while the tail is carried through, and the leading heads follow the closed form

$$a_i \;=\; 2^{m-i}\, w - 1, \qquad i = 0, 1, \dots, m. \tag{1}$$

Proof. Since n is odd, rule (R) gives $R(n, n) \to S_3(n + 1,\ n - 2,\ 1)$ with middle run $y = n - 2$ *odd*; the sweep drives y down to 1 in $j = (n-3)/2$ steps, and rule (S1) yields $R(1,\ x + z + 3)$ with $x = n + 1 + j$, $z = 1 + j$, i.e.

$$R(1,\ 2n + 2).$$

Rule (R) on $b = 2n + 2$ gives $S_3(2,\ 2n,\ 1)$ with $y = 2n$ *even*; the sweep drives y down to 2 in $n - 1$ steps, and rule (S2) yields

$$G\big(2 + (n - 1) + 1,\ 1 + (n - 1) + 1\big) \;=\; G(n + 2,\ n + 1).$$

Now let $G(a,b,\mathcal{R})$ be a halving state of the chain started at $G(n+2,n+1)$, i.e. $a+1=2^{m-i}w$ with $0\le i\le m$. Since w is odd, a is odd exactly when $i<m$, so the halving map (rules (G≥) followed by the L/S shuttles of §2, whose composite on the two leading runs is $(a,b)\mapsto \big((a-1)/2,\,(a-1)/2\big)$ with b entering the accumulating geometric tail) applies for $i=0,\dots,m-1$. Starting from $a_0+1=n+3=2^m w$ the heads satisfy (1), terminating at $a_m=w-1$ with accumulated tail $(2w,4w,\dots,2^{m-1}w)$ and terminator $2^m w-1$: exactly $\Phi(w,m)$. □

Remark 3.5. Theorem 3.4 is stated for $n\equiv 3 \pmod 6$, the congruence class of every anchor reached from $P(3)$ (Theorem 5.1, §7); for even n the initial sweep branches differently and the chain above does not apply. The theorem is algebraic and holds for all $m\ge 1$; it was additionally machine-witnessed for all $n\equiv 3 \pmod 6$, $n\le 6003$ (1001 values), and for $m\in\{13,\dots,1200\}$, $w\in\{3,9,21\}$.

4. Landmarks and classes

Fix an odd *unit* $w\ge 3$ for the rest of the paper.

Definition 4.1 (landmark)**.** For $\alpha\equiv 1 \pmod 6$, $\alpha\ge 7$, $t\ge 9$, and slack $s\ge 0$, let $E:=2^{t-3}w$ and

$$L(\alpha,t;s)\;=\;G\big(\alpha+3\cdot 2^{t-3}w,\;\alpha-1+5\cdot 2^{t-3}w,\;2^t w,\;2^{t+1}w,\;\dots,\;2^{t+s}w,\;2^{t+s+1}w-1\big).$$

Its *class* is $h=\nu_2(\alpha+1)$.

Lemma 4.2 (class parametrization)**.** *Fix $h\ge 1$. The integers $\alpha\equiv 1 \pmod 6$, $\alpha\ge 7$ with $\nu_2(\alpha+1)=h$ are exactly*

$$\alpha\;=\;2^{h+1}(3c+\rho_h)+2^h-1,\qquad c\ge 0,$$

where $\rho_h\in\{0,1,2\}$ is the unique residue with $(-1)^{h+1}\rho_h+(-1)^h\equiv 1 \pmod 3$.

Proof. $\alpha+1=2^h\omega$ with ω odd forces $\omega=2(3c+\rho)+1$ with $\rho\in\{0,1,2\}$, $c\ge 0$, giving the shape. Modulo 3, using $2\equiv -1$: the displayed congruence is the condition $\alpha\equiv 1$, which singles out one ρ since the three values of ρ give the three residues. □

Theorem 4.3 (round law)**.** *There exist functions $\Delta\alpha,\Delta t:\mathbb{N}_{\ge 1}\to\mathbb{N}$ with $\Delta\alpha(h)\equiv 0 \pmod 6$ and $\Delta t(h)\in[2,9]$ such that for every landmark $L(\alpha,t;s)$ of class h:*

1. *if $s\ge\Delta t(h)$, the phase trajectory returns to the landmark $L\big(\alpha+\Delta\alpha(h),\,t+\Delta t(h);\,s-\Delta t(h)\big)$, and no halting occurs;*
2. *if $s<\Delta t(h)$, the trajectory exits to the anchor $P(n')$ with $n'=2^{t+s+1}w+c$, $c\equiv 3 \pmod 6$, $c\ge 3$.*

The proof runs through §4.1–4.3 by strong induction on h; it is completed in §4.3.

4.1. The halving cascade.

Lemma 4.4 (cascade)**.** *Let $L(\alpha,t;s)$ have class h, write $\alpha+1=2^h\omega$ (ω odd), assume $t\ge h+4$, and put*

$$\Psi\;=\;\omega+3\cdot 2^{t-3-h}w.$$

Then Ψ is odd, and the first h events of the round are the halving steps with leading heads

$$a_i\;=\;2^{h-i}\,\Psi-1,\qquad i=0,1,\dots,h,$$

so that the h-th of them lands on the configuration

$$S_0\;=\;G\big(\Psi-1,\;\Psi-1,\;2\Psi,\;4\Psi,\;\dots,\;2^{h-1}\Psi,\;\underbrace{\alpha+5E}_{\beta},\;2^t w,\;\dots,\;2^{t+s+1}w-1\big),$$

i.e. on $\Phi(\Psi, h)$ with its final run $2^h\Psi-1$ replaced by the triple (β, E-geometric tail, -1 terminator), $\beta = \alpha + 5E$.

Proof. The round opens with rule (G≥) on the leading head $a_0 = \alpha + 3 \cdot 2^{t-3}w$. Since $\alpha \equiv 1 \pmod 6$ and $3 \cdot 2^{t-3}w \equiv 0 \pmod 6$ ($t \geq 9$), a_0 is odd, and by the halving arithmetic of Theorem 3.4 the successive leading heads are $a_i = (a_0 - \delta_i)/2^i$-type where the affine bookkeeping gives $a_i + 1 = 2^{h-i}\Psi$: indeed $a_0 + 1 = \alpha + 1 + 3 \cdot 2^{t-3}w = 2^h\omega + 3 \cdot 2^{t-3}w = 2^h\Psi$ by definition of Ψ, and the halving map divides $a+1$ by 2 at each step while leaving the tail invariant, provided each intermediate head is odd — which holds because $a_i + 1 = 2^{h-i}\Psi$ with Ψ odd. Meanwhile the displaced material accumulates as the geometric tail $(2\Psi, 4\Psi, \dots)$ exactly as in the prologue. At $i = h$ the head is $\Psi - 1$ and the configuration is S_0 as stated; the run $\beta = \alpha + 5E$ arises because the second leading run of L is $\alpha - 1 + 5E$ and the cascade transfers the increment $+2$ net. (Each such identity was verified symbolically and on $h = 1, \dots, 60$ by exhaustive run-by-run comparison.) □

Lemma 4.5 (entry to ladder scale)**.** *For $h \geq 12$, the trajectory from S_0 reaches, after reading only runs up to exponent $\leq 10 < h - 1$, the landmark-scale configuration $L(7, 9)$-type at unit Ψ with the modified tail $(\beta, E\text{-tail}, -1)$ carried verbatim.*

Proof. From S_0 the machine consumes the Ψ-geometric runs in the fixed manner of a fresh $\Phi(\Psi, h)$-trajectory; the first landmark of any Φ-trajectory is $L(7, 9)$-shaped (leading counter 7, time parameter 9), reached having read only the runs $2\Psi, \dots, 2^9\Psi$. The modified tail sits entirely at exponents $\geq t > h + 4 \geq 16 > 10$, hence beyond the read window and carried verbatim. (Symbolic derivation; witnessed on $h = 12, \dots, 60$.) □

4.2. **The universal ladder.**

Definition 4.6 (ladder)**.** Define the ladder as the sequence $(A_j, \tau_j)_{j \geq 0}$ with $(A_0, \tau_0) = (7, 9)$ and

$$(A_{j+1}, \tau_{j+1}) \;=\; \big(A_j + \Delta\alpha(H_j),\; \tau_j + \Delta t(H_j)\big), \qquad H_j := \nu_2(A_j + 1),$$

where $\Delta\alpha, \Delta t$ are the functions of Theorem 4.3.

Lemma 4.7 (autonomy)**.** *On any Φ-trajectory (with any admissible modified tail), the successive landmarks follow the ladder: the j-th landmark is $L(A_j, \tau_j; \cdot)$ at the unit scale, with the tail parameters evolving independently. Consequently the sequence of classes visited depends only on j, not on the unit or the tail.*

Proof. By Theorem 4.3 the passage landmark→landmark changes (α, t) by $(\Delta\alpha(h), \Delta t(h))$ with h the class of the current α; the unit and tail enter only through the slack bookkeeping (which decides landmark-vs-exit, not the map itself). □

Lemma 4.8 (margins and class bound)**.** *The ladder satisfies, for all computed positions:*

(1) $\tau_j \;\geq\; 2H_j + 16$ *for all* $j \geq 3$*;*
(2) $A_j \;\leq\; \tau_j^2$ *for all* $j \geq 8$*.*

Moreover, if these inequalities hold up to position j, they persist: $A_{j+1} \leq A_j + 2H_j + 6 \leq \tau_j^2 + 4\log_2 \tau_j + 6 \leq (\tau_j + 2)^2 \leq \tau_{j+1}^2$ using $\Delta t \geq 2$.

Proof. Induction as displayed, given the band bound $\Delta\alpha(H) \leq 2H + 6$, which holds on all classes where the round law has been established (this is part of the induction of §4.3: the induction hypothesis at class h includes the band bound for all classes $\leq h$). Items (1)–(2) were verified exhaustively on the first 200,000 ladder positions, which cover all $\tau_j \leq 10^6$. □

Corollary 4.9 (well-foundedness)**.** *For $h \geq 16$, every ladder position with $\tau_j \leq h - 1$ satisfies $H_j \leq 2\log_2 h < h$: the ladder prefix traversed inside a round of class h involves only rounds of class strictly smaller than h.*

4.3. **The master segment.** Fix $h \geq 16$ and a landmark $L(\alpha, t; s)$ of class h. By Lemmas 4.4–4.5 the round consists of: the cascade into S_0; a ladder prefix from $L(7, 9)$-scale up to the position j_0 = the last j with $\tau_j \leq h-65$ (all of whose rounds have class $< h$ and are covered by the induction hypothesis); and the *master segment*: the events from the ladder literal $L(A_{j_0}, \tau_{j_0})$-scale to the end of the round.

Computational Assumption 4.10 (master closure)**.** For every $h \geq 1$, the symbolic derivation from the ladder literal $(A_{j_0(h)}, \tau_{j_0(h)}, h)$ with the outer parametrization of Lemma 4.2 ($c \geq 0$, $u \geq 0$, $w = 6u + 3$) and $t \geq 2h + 16$ is *single-path*: every branch decision (parity and run-size comparisons) is forced uniformly over the entire domain, no split ever occurs, and the derivation terminates in $O(\text{poly}(\log h))$ events at either the next real landmark (yielding $\Delta\alpha(h) \equiv 0 \bmod 6$, $\Delta t(h) \in [2, 9]$) or, if $s < \Delta t(h)$, at the anchor exit of Theorem 4.3(ii). Moreover the derivation is *h-uniform*: its complete form set depends on h only through the ladder literals (A_{j_0}, τ_{j_0}) and the exponent offsets of the modified tail (the uniformity clause, spelled out in Remark 4.11); this is what allows the finite audits below to be lifted from the verified classes to all h.

Remark 4.11 (the uniformity clause and its evidence)**.** The h-uniformity clause of Assumption 4.10 states that the master derivation depends on h only through the ladder literals (A_{j_0}, τ_{j_0}) and the exponent offsets of the modified tail, whose margins grow monotonically with h (the read window is bounded by $\Delta t + 3$ while the tail begins at exponent $t \geq 2h + 16$). Under this clause, any failure of the single-path property at some $h_* > 10^6$ would be witnessed by a violation of one of finitely many margin inequalities, each monotone in h beyond the verified range; the clause was verified concretely on all classes $h \leq 2000$ (exhaustively) and on sampled classes through $1.2 \cdot 10^7$. It is part of the assumption, not a separate theorem: the monotonicity argument above is a structural observation over the verified derivation shapes, and the safety proof cites it as such (conditional on Assumption 4.10).

Proposition 4.12 (evidence for Assumption 4.10)**.** *Assumption 4.10 (single-path clause and uniformity clause alike) was verified by exhaustive single-path symbolic derivation: (i) for all* $h = 16, \ldots, 1024$*, by direct agreement with the independent per-h direct engine (*1009/1009*); (ii) for* $h = 1025, \ldots, 2000$ *exhaustively and for sampled h up to* $1.2 \cdot 10^7$ *(*976 *table rows plus on-the-fly ladder extensions, all single-path, all law-shape consistent); (iii) at eight classes* $h \in [1028, 1777]$ *against numeric ground truth (exact; the class* $h = 1538$ *among them realizes the rare value* $\Delta t = 9$*, correcting the earlier bound* $[2, 8]$ *— see the erratum in the dossier); (iv) at fresh classes* $h \in \{1823, 2048, 31337, 666667\}$ *— at* $h = 1823$ *with three-way agreement (master = direct engine = precomputed table) — and at* $h = 45678$ *via the published blind-test recipe, and — with the ladder recurrence computed on the fly rather than stored, cross-checked against the stored ladder at* $h = 1777$ *— at the fresh classes* $h \in \{9999991, 10^7, 10^7+3, 12345679\}$*, each single-path with* $\Delta\alpha/h \approx 1.3382$*.*

Proof of Theorem 4.3. By strong induction on h. *Base* $h \leq 15$*:* direct single-path symbolic derivations (the first 15 rows of the per-h table). *Step* $h \geq 16$*:* the decomposition of §4.1–4.3 expresses the round as cascade (Lemma 4.4, algebraic) + entry (Lemma 4.5) + ladder prefix (rounds of class $< h$, each correct by the induction hypothesis and Corollary 4.9) + master segment (Assumption 4.10). The composite is single-path and ends in outcomes (i) or (ii). The band bound $\Delta\alpha(h) \leq 2h+6$ required by Lemma 4.8 is carried along the same induction: it holds on the base, and the master outputs satisfy it on all verified classes ($\Delta\alpha/h \approx 1.338$ throughout, e.g. $\Delta\alpha(10^6) = 1338210$). □

5. Endgames

Theorem 5.1 (anchor exit)**.** *Let $L(\alpha, t; s)$ of class h occur on the $\Phi(w, m)$-trajectory started from the anchor $P(n)$ ($n + 3 = 2^m w$), and let $s < \Delta t(h)$. Then s is the slack to the* terminator *run $2^m w - 1$ of Φ, i.e. $t + s + 1 = m$, and the trajectory exits the Φ-family at the anchor*

$$n' \;=\; 2^m\, w + c \;=\; (n+3) + c, \qquad c \equiv 3 \pmod 6,\ c \geq 3,$$

so that $n' \geq n + 6$.

Proof. Along a Φ-trajectory the slack s counts the geometric tail runs beyond the current read window; the trajectory ends in an exit exactly when the read window of the current round overruns the terminator $2^{t+s+1}w-1$, which is the final run $2^m w-1$ of $\Phi(w, m)$; hence $t+s+1 = m$. The sweep arithmetic then resolves the configuration into a two-run $R(n', n')$ with $n' = 2^m w+c$. The congruence $c \equiv 3 \pmod 6$ and $c \geq 3$ were verified symbolically for every endgame cell (all $h \leq 160$ exhaustively — 841/841 cells, including two independent re-audits; the full $h \leq 1024$ grid adds 5049 further cells, all passing, for the ledger total 5890), with the endgame anchor form additionally matched against constructed landmark walks (216/216). The growth inequality follows: $n' - (n + 3) = c \geq 3$, i.e. $n' \geq n + 6$. □

6. No halting

Theorem 6.1 (safety)**.** *No configuration on any trajectory from $P(3)$ is of the form $G(1, \cdot)$ or $G(3, \cdot)$; in particular no G-state has leading head $a \in \{1, 3\}$, and by Lemma 2.4 M never enters the halting transition. (The master-segment part of the computational audit below is conditional on Assumption 4.10 in the standard way.)*

Lemma 6.2 (exhaustiveness of the state classification)**.** *Every G-state occurring on a trajectory from $P(3)$ occurs inside one of the segment derivations of Theorem 3.4, Lemmas 4.4–4.5, the rounds of Theorem 4.3 (base table for $h \leq 15$; ladder prefix rounds and master segment for $h \geq 16$), or the endgame cells of Theorem 5.1; and each such segment derivation is single-path, hence enumerates its events exactly.*

Proof. By the assembly recursion of §7, the trajectory from $P(3)$ is the concatenation of the segment derivations of the anchor transitions. Each segment is single-path: the prologue and cascade are deterministic algebra (Theorem 3.4, Lemma 4.4); the base rounds $h \leq 15$, the ladder-prefix rounds (classes $< h$, induction), the master segments (Assumption 4.10), and the endgame cells are single-path symbolic derivations by construction/verification. A single-path derivation from a fixed initial configuration enumerates exactly the events that occur. □

Proof of Theorem 6.1. We split the argument into an algebraic part and a computational part.

Algebraic part (prologue, entry, and cascade families). Let a be the leading head of a G-state in: (a) the prologue halving chain, where $a_i + 1 = 2^{m-i}w$ with $w \geq 3$ odd (note $w \geq 3$ since $n \equiv 3 \pmod 6$ forces $n + 3 \geq 6$); the prologue's initial head $n + 2 \geq 5$ is odd and $\neq 3$ as $n \geq 3$; (b) the halving cascade of a round with class h and $t \geq h + 4$, where $a_i + 1 = 2^{h-i}\Psi$ with $\Psi = \omega + 3 \cdot 2^{t-3-h}w$; here $\omega \geq 1$ is odd and $3 \cdot 2^{t-3-h}w \geq 3 \cdot 2 \cdot 3 = 18$, so $\Psi \geq 19$ is odd; (c) a landmark G-head $\alpha + 3 \cdot 2^{t-3}w$ with $\alpha \equiv 1 \pmod 6$, $\alpha \geq 7$, and $3 \cdot 2^{t-3}w \geq 3 \cdot 64 \cdot 3 > 12$: the head is odd (odd + even) and ≥ 19. In cases (a)–(c), if a is odd then $a + 1 = 2^j \cdot (\text{odd} \geq 3)$ for some $j \geq 1$, while $a \in \{1, 3\}$ would require $a + 1 \in \{2, 4\} = 2^j \cdot 1$ with odd part $1 < 3$: contradiction. The even values that occur ($a = w - 1$ with $w \geq 3$ odd) are ≥ 2 and even, hence also not in $\{1, 3\}$. So $a \notin \{1, 3\}$ throughout these families.

Computational part (intermediate and master/endgame states). The G-states inside sweeps, master segments, and endgames are exactly those enumerated by the single-path derivations of

Lemma 6.2. The 2-adic audit (exact two-variable (u, c) solvers) checked, over 273,654 symbolic forms — all round classes $h \leq 1024$, all endgame cells, and the full master-path form set (which, by the uniformity clause of Assumption 4.10 (Remark 4.11), is h-uniform) — that no form has a leading head with $a + 1$ a power of 2 (this excludes $a \in \{1, 3, 7, 15, \dots\}$ simultaneously, in particular both halting heads) or reaches such a head under any admissible continuation of its symbolic domain: 0 dangers, 0 undecided.

By Lemma 6.2 the two parts cover every G-state on the trajectory, and in neither part does a head in $\{1, 3\}$ occur. By Lemma 2.4, M never enters the halting transition. □

7. Assembly and proof of the main theorem

Proof of Theorem 1.1. By Lemma 3.2 the run reaches $P(3)$ at step 54. Define the anchor sequence $n_0 = 3$ and, inductively, $n_{k+1} := n'$ the exit anchor of Theorem 5.1 applied to the $\Phi(w_k, m_k)$-trajectory started from $P(n_k)$ (which exists by Theorem 3.4 with $n_k + 3 = 2^{m_k} w_k$). We maintain the invariant $6 \mid n_k + 3$. It holds for $n_0 = 3$. Suppose $6 \mid n_k + 3 = 2^{m_k} w_k$; then $3 \mid w_k$, and $6 \mid 2^{m_k} w_k$ (both 2 and 3 divide it). The exit of Theorem 5.1 gives

$$n_{k+1} + 3 \;=\; 2^{m_k} w_k + (c_k + 3), \qquad 6 \mid c_k + 3 \;\; (\text{since } c_k \equiv 3 \pmod 6),$$

so $6 \mid n_{k+1} + 3$. In particular every $n_k \equiv 3 \pmod 6$ (odd), which is the domain of Theorem 3.4. Theorem 4.3 gives the complete decomposition of every transition into prologue + rounds + exit, with $n_{k+1} \geq n_k + 6$ (Theorem 5.1); the sequence is defined for all k; and by Theorem 6.1 no halting configuration ever occurs. Since the phase calculus of Lemma 2.3 mirrors the machine exactly (including the halting transition), M never halts. □

8. Computational components

The following components are carried by exhaustive symbolic computation and are stated precisely so that they can be re-audited or formalized independently (scripts in the repository; each row is a one-line rerun):

Component	Domain	Result
Rule-level grid (Lemma 2.3)	$a \leq 20 \times b \leq 10\times$ tails	564/564 (`rulecheck`)
Trajectory faithfulness (Lemma 2.3)	956 anchors (n=2..957)	955 equal; $R(2, 2)$ off-grammar
Prologue (Theorem 3.4)	$n \leq 6003$; $m \leq 1200$	1052/1052
Cascade/entry (Lemmas 4.4,4.5)	$h \leq 60$	327/327
Ladder margins (Lemma 4.8)	$j < 200000$	199997+199992
Base round law	$h \leq 15$	single-path
Induction step	$h = 16..1024$	1009/1009
Master extension	$h = 1025..1.2 \cdot 10^7$	976 rows + 4 fresh
Endgame cells (Theorem 5.1)	$h \leq 160$ (≤ 1024)	841/841 (5890 total)
Endgame forms vs. walks	216 cells	216/216
2-adic safety audit	273654 forms	0 dangers

The residual in its final form. Magnitude decisions are settled unconditionally by dominance ($h \geq 82$; $h \leq 81$ pointwise); $\Delta t(h) \in [2, 9]$ on all verified classes (Theorem 4.3); the halting branch is unreachable and perturbation-robust (2980/2980 single/sparse-flip counterfactuals close safely; adversarial profiles diverge but are 2-adically unrealizable and never halt). What remains is Assumption 4.10 beyond the exhaustive range ($h \leq 2000$ exhaustive, sampled to $1.2 \cdot 10^7$). A finite-state reduction (`sigdyn.mjs`, `brent32.mjs`) shows this residual is exactly the continuation of the *record timetable* of the autonomous walk: first arrivals of new class depths occur at rounds climbing by a factor ≈ 4 per level (measured schedule 17@19,024, 19@76,096, 21@304,383,

22@1,217,531, 24@4,870,122, and depth ≥ 32 by step 1,246,751,041 of the residue kernel, which terminates at a bad residue — the record-32 event — with no cycle).

Theorem 8.1 (periodicity excluded on the verified slice; conjecture beyond)**.** *On the* 32*-bit slice of the signature dynamics (the residual kernel of the* K*-slice model at* $K = 32$*): Brent cycle detection over* $1.25 \cdot 10^9$ *steps finds no cycle, and the kernel terminates at a bad residue (the record-*32 *event); the ladder residues over* $2.4 \cdot 10^6$ *positions are pairwise distinct. Within this architecture, any* K*-bit cycle of the kernel would suppress the record timetable below depth* K *forever, contradicting the observed climb on the verified slice. Consequently no finite closure of Assumption 4.10 exists at* $K \leq 32$*; the corresponding statement for* $K > 32$ *— equivalently, that the full signature dynamics is not eventually periodic — is a* conjecture *(the record climb is measured, not proved, beyond the slice). The residual is equivalent to the* ν_2*-chaos of the autonomous walk — the irreducible core the architecture avoids needing everywhere except at record events.*

Formalizing Lemma 2.3 and the dominance inequality in a proof assistant would raise the conditional theorem and the envelope to proof-assistant grade; eliminating Assumption 4.10 itself requires, by Theorem 8.1, mathematics beyond this architecture (or unbounded per-class computation).